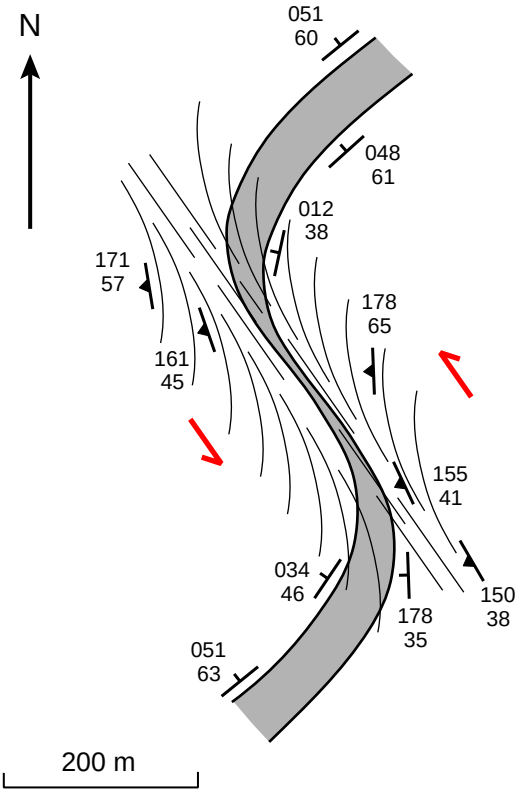
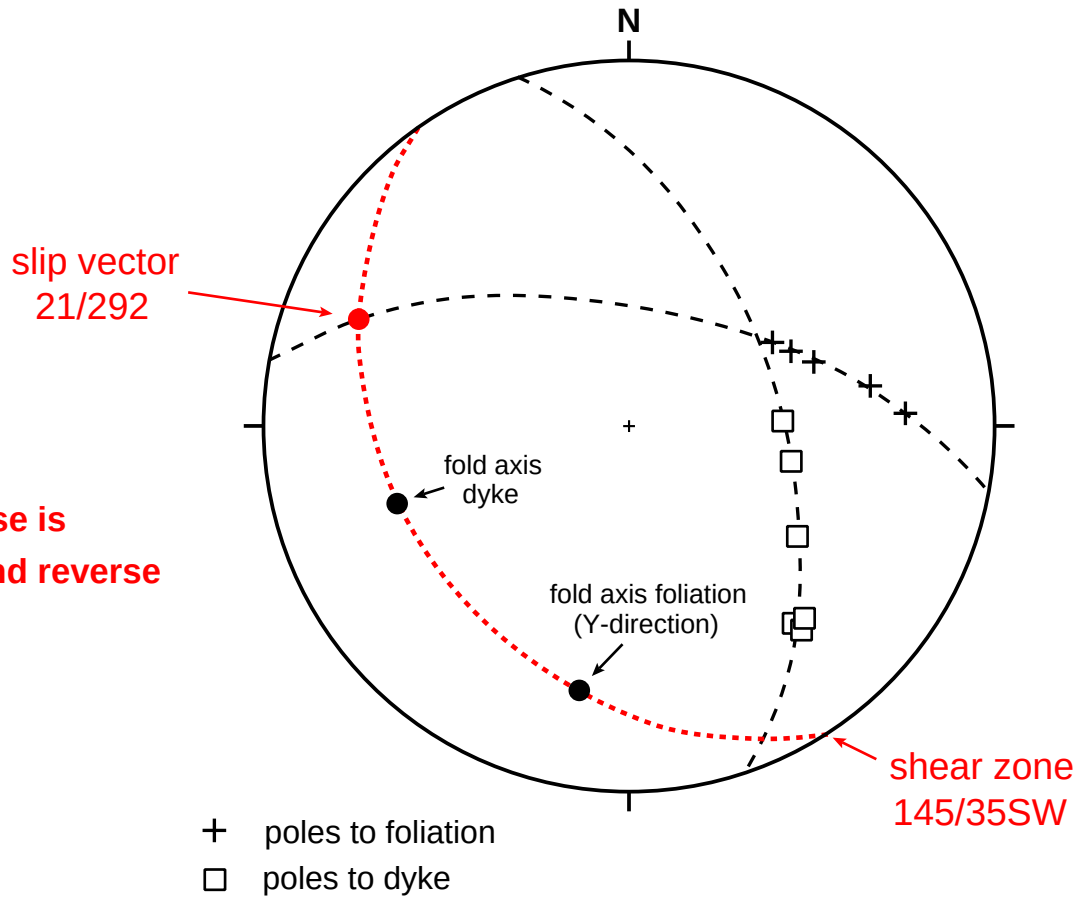
SHEAR ZONE EXERCISE

The map below shows the outcrop of a foliated plastic shear zone on a horizontal surface. The shear zone offsets a dyke. Orientation data are given for the dyke as it curves into the shear zone, and for the foliation



Determine the orientation of the shear zone and slip vector, the sense of shear, and the magnitude of the slip vector

1. Plot poles to dyke; the pole to the profile plane so defined lies in the shear plane
2. Plot poles to foliation; the pole to the profile plane so defined lies in the shear plane at 90° to the slip vector (it is the Y-direction)
3. Draw in the shear plane and check that its strike agrees with the map trend of the shear zone; count around 90° on the shear plane from the Y-direction and this is the slip vector
4. Determine sense of shear from orientation of slip vector and shear plane, noting that the sense of strike-slip follows from the offset of the dyke on the map



Shear sense is sinistral and reverse